



Q1: If an event A satisfies $P(A) = 0.3$ then $P(\bar{A} | \Omega)$ equals
 A) 0.3 B) 0.7 C) 0.4 D) 0.6 E) None

Q2: For some independent events A, B with $P(A) = P(B) = 0.2$ we have $P(A \cap B)$ equals
 A) 0.01 B) 0.1 C) 0.2 D) 0.04 E) None

Q3: Let X be a random variable with zero mean. We have $E(2X^2 + X) =$
 A) $2 \text{Var}(X)$ B) $3 \text{Var}(X)$ C) $9 \text{Var}(X)$ D) $4 \text{Var}(X)$ E) None

Q4: For every $x \leq 0$ we have $2u(-x) + 3\delta(x) =$
 A) 0 B) 2 C) undefined D) 1 E) None

Q5: Each ordered pair in $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$ has equal probability. The expected value of XY equals
 A) 0.1 B) 0.2 C) 0.25 D) 0.3 E) None

Q6: Let X takes values 0 and 1 with $P(X=0) = 0.9$ and let Y takes values 1 and 2 with $P(Y=2) = 0.8$. If X and Y are independent, then $f(1,1) =$
 A) 0.1 B) 0.01 C) 0.2 D) 0.02 E) None

Q7: For some continuous function f if $\int_{-1}^1 f(x) \delta(x) dx = 0$, then $f(0) =$
 A) 0 B) 1 C) -1 D) -2 E) None

Q8: For the joint PDF $f(x,y) = y$ when $0 < x < 1$, $0 < y < \sqrt{2}$ and zero otherwise. We have $f_Y(y) =$
 A) 1 B) y C) $\sqrt{2}y$ D) x E) None

Q9: For continuous random variable X with $P(1 < X < 2) = 0.1$, Then $P(0 < X < 2) - P(0 < X < 1)$ equals
 A) 0.3 B) 0.4 C) 0.2 D) 0.1 E) None

Q10: For two random variables X, Y with $\text{Cov}(X, Y) = 1$, $\text{Cov}(\pi + eX, e + \pi Y) =$
 A) $-e\pi$ B) e C) $e\pi$ D) π E) none

$(2) - P(0) - P(1)$

$P(1) - P(0)$
 $0.1 - 0$

$9 - 0.1$

$P(2) - P(1) = 1$
 $P(2) =$



Q11: Let X, Y be two independent random variables with $\text{Var}(X) = \text{Var}(Y)$. Then $\text{Corr}(X, Y) =$

A) 0

B) 1

C) 2

D) 3

E) None

Q12: A discrete random variable X with CDF

$$F(x) = a u(x+2) + 3a u(x-1)$$

Then the constant a equals

A) 0

B) 0.25

C) 0.2

D) 0.5

E) None

a1

=1

$$3a + 1 = \frac{1}{4}$$

$$a + 3a$$

$$4a = 1$$